

A Geometric Correction-Series Derivation of the Sagnac Return-Time Difference

Carmen N. Wrede, Lino P. Casu, Bingsi

We present a short kinematic derivation of the Sagnac return-time difference using an iterative geometric correction series. Instead of starting from holonomy or synchronization arguments, the travel times of the co- and counter-propagating beams are reconstructed as geometric series. Their limits reproduce the standard Sagnac time delay and its continuous loop-integral form. This shows that the oscillatory zig-zag structure of the correction sequence is not a physical oscillation of light, but a convergence pattern of the iterative calculation. The later phase difference is therefore interpreted as an interference-language readout of the accumulated arrival-time/path difference, not as the mechanism represented by the correction series itself.

1. Iterative Geometric Correction-Series Derivation

There is a very simple way to see the Sagnac return time difference without invoking holonomy at all. It is essentially the same type of iterative correction that appears in elementary pursuit problems.

Consider a light signal propagating along a path of length L , while the receiver moves with speed v . All quantities in the following derivation are initially defined in the non-rotating inertial axle frame. L denotes the circular path length measured in that frame, v is the tangential speed of the moving detector in that frame, and $t_{\pm}$ denote coordinate return times in that frame.

For the co-rotating direction, the receiver moves away during the light travel time. A first estimate is:

$$t_0 = \frac{L}{c}$$

But during this time the receiver has moved an additional distance:

$$v \cdot t_0 = v \cdot \frac{L}{c}$$

So the light must cover an extra distance. This produces a correction proportional to v/c . Repeating the same argument gives:

$$t_+ = \frac{L}{c} \cdot \left[1 + \frac{v}{c} + \left(\frac{v}{c}\right)^2 + \left(\frac{v}{c}\right)^3 + \dots \right]$$

Since $v < c$, this geometric series converges:

$$t_+ = \frac{L}{c} \cdot \frac{1}{1 - \frac{v}{c}}$$

and therefore:

$$t_+ = \frac{L}{c - v}$$

For the counter-propagating direction, the receiver moves toward the light signal. The first estimate $\frac{L}{c}$ is therefore too large. The first correction subtracts the distance the receiver has moved toward the light. But this subtraction slightly overcorrects, so the next term must be added back, and so on. This gives the alternating correction series:

$$t_- = \frac{L}{c} \cdot [1 - \frac{v}{c} + (\frac{v}{c})^2 - (\frac{v}{c})^3 + \dots]$$

This converges to:

$$t_- = \frac{L}{c} \cdot \frac{1}{1 + \frac{v}{c}}$$

and therefore:

$$t_- = \frac{L}{c + v}$$

Thus the return time difference is:

$$\Delta t_{axle} = t_+ - t_-$$

$$\Delta t_{axle} = \frac{L}{c - v} - \frac{L}{c + v}$$

$$\Delta t_{axle} = \frac{2Lv}{c^2 - v^2} \approx \frac{2Lv}{c^2}$$

For a circular path with:

$$L = 2\pi R$$

$$v = \Omega R$$

$$A = \pi R^2$$

this becomes:

$$\Delta t_{axle} = \frac{4A\Omega}{c^2 - v^2}$$

and for v much smaller than c:

$$\Delta t_{axle} \approx \frac{4A\Omega}{c^2}$$

This is the usual Sagnac result in the small-v/c limit.

The successive correction terms form a damped zig-zag convergence pattern when displayed graphically. This pattern is not a physical oscillation of light: the light does not move back and forth, nor does it locally slow down or accelerate. The rotation of the platform generates the nonzero return-time asymmetry, while the zig-zag form belongs only to the iterative mathematical reconstruction of that asymmetry.

2. From Correction Series to Loop Integral

The loop-integral description and the iterative correction-series description are two equivalent decompositions of the same continuous return time. In the loop-integral representation, one integrates once around the fixed ring while accounting for the moving closure point through the relative closing speed $c \mp v$. In the correction-series representation, one reconstructs the same return time through successive signed residual-distance corrections generated by the motion of the detector.

The iterative geometric correction series: The continuous problem is broken down into many small correction steps. Each individual step is just a partial contribution, or a residual error.

The continuous description: When the limit is taken, the individual successive correction terms lose their independent significance. They merge into a single aggregate value, the return time difference.

$$\oint \frac{1}{c-v} ds = \frac{L}{c-v} = \lim_{N \rightarrow \infty} \sum_{n=0}^N \frac{L}{c} \cdot \left(\frac{v}{c}\right)^n$$

$$\Delta t = \oint \left[\frac{1}{c-v} - \frac{1}{c+v} \right] ds = \frac{2Lv}{c^2 - v^2} = \lim_{N \rightarrow \infty} \sum_{n=0}^N \frac{L}{c} \cdot \left[\left(\frac{v}{c}\right)^n - \left(-\frac{v}{c}\right)^n \right]$$

The geometric correction series is the β -series expansion of the continuous loop-integral integrand, evaluated around the closed path for constant β . The observed mathematical oscillation does not represent a physical property of the light itself, which propagates at a constant speed c , but rather reconstructs the geometry-dependent return-time asymmetry. The two propagation directions generate correction series related by the substitution $\beta \rightarrow -\beta$. Their difference isolates the odd, direction-dependent component that constitutes the Sagnac return-time asymmetry.

Therefore, no holonomy-based mechanism is required to derive the observed Sagnac return-time difference. The correction series represents the kinematic reconstruction of the return-time asymmetry generated by the moving closure point. A separate synchronization-gap description may be introduced additionally, but it is not the mechanism represented by this series.

Defining

$$\beta = \frac{v}{c}, T_0 = \frac{L}{c}$$

The two directed return-time series may be written as

$$t_+ = T_0 \sum_{n=0}^{\infty} \beta^n, \quad t_- = T_0 \sum_{n=0}^{\infty} (-\beta)^n$$

Their sum isolates the even, direction-independent contribution:

$$t_+ + t_- = 2T_0(1 + \beta^2 + \beta^4 + \dots)$$

whereas their difference isolates the odd, direction-dependent Sagnac contribution:

$$t_+ - t_- = 2T_0(\beta + \beta^3 + \beta^5 + \dots)$$

Consequently, the normalised return-time asymmetry is exactly

$$\boxed{\frac{t_+ - t_-}{t_+ + t_-} = \beta = \frac{v}{c}}$$

Thus the directed Sagnac signal is the odd component of two return-time series related by direction reversal, while the even component is shared by both directions.

3. Coordinate Return Time and Proper-Time Readout of the Rotating Detector

The geometric correction-series derivation above is formulated in the non-rotating inertial axle frame. In this frame, the circular optical path has circumference

$$L = 2\pi R$$

the detector moves tangentially with speed

$$v = \Omega R, \quad \beta = \frac{v}{c} = \frac{\Omega R}{c}$$

and the two directed return times are coordinate times of the axle frame:

$$t_+ = \frac{L}{c-v}, \quad t_- = \frac{L}{c+v}$$

Their difference is therefore

$$\Delta t_{\text{axel}} = t_+ - t_- = \frac{2Lv}{c^2 - v^2}$$

For a circular path with $L = 2\pi R$, $A = \pi R^2$, $v = \Omega R$, this becomes

$$\Delta t_{\text{axle}} = \frac{4A\Omega}{c^2 - \Omega^2 R^2}$$

Introducing the dimensionless parameter

$$\beta = \frac{v}{c}$$

the same expression can be written as

$$\Delta t_{\text{axle}} = \frac{4A\Omega}{c^2} \frac{1}{1 - \beta^2}$$

Since

$$\frac{1}{1-\beta^2} = 1 + \beta^2 + \beta^4 + \beta^6 + \dots = \gamma^2$$

with

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

the geometric correction-series result may equivalently be written as

$$\Delta t_{\text{axle}} = \frac{4A\Omega}{c^2} (1 + \beta^2 + \beta^4 + \beta^6 + \dots) = \frac{4A\Omega}{c^2} \gamma^2.$$

This expression is the return-time difference expressed in the coordinate time of the non-rotating inertial axle frame. It must not be confused with the proper-time difference registered by the detector mounted on the rotating platform.

The two counter-propagating signals return to the worldline of the same moving detector at two distinct reception events. The clock carried by this detector measures proper time. The clock carried by this detector measures proper time. In flat spacetime, its proper time is related to the coordinate time of the axle frame by

$$d\tau_{\text{det}} = dt_{\text{axle}} \sqrt{1-\beta^2} = \frac{dt_{\text{axle}}}{\gamma}$$

Therefore, the return-time difference recorded by the detector's own clock is

$$\Delta\tau_{\text{det}} = \Delta t_{\text{axle}} \sqrt{1-\beta^2}$$

Substituting the geometric correction-series expression gives

$$\Delta\tau_{\text{det}} = \frac{4A\Omega}{c^2} (1 + \beta^2 + \beta^4 + \beta^6 + \dots) \sqrt{1-\beta^2}$$

This product form makes the separation of the two ingredients explicit. The geometric series reconstructs the Sagnac return-time asymmetry in the axle frame, while the factor

$$\sqrt{1-\beta^2} = \frac{1}{\gamma}$$

converts that coordinate-time difference into the proper-time difference of the rotating detector.

Using

$$\gamma^2 \sqrt{1 - \beta^2} = \gamma$$

the detector proper-time result reduces to

$$\Delta\tau_{\text{det}} = \frac{4A\Omega}{c^2} \gamma$$

The corresponding Taylor expansion is therefore

$$\Delta\tau_{\text{det}} = \frac{4A\Omega}{c^2} \left[1 + \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + \frac{5}{16}\beta^6 + \dots \right]$$

This series is proportional to γ , not to γ^3 . A dependence proportional to γ^3 would arise only if the axle-frame coordinate-time result, which already contains the factor γ^2 , were incorrectly multiplied by an additional Lorentz factor γ .

For the proper-time readout of the co-rotating detector, the conversion proceeds in the opposite direction: the coordinate-time result must be multiplied by

$$\sqrt{1 - \beta^2} = \frac{1}{\gamma}$$

The two exact flat-spacetime quantities must therefore be kept distinct:

$$\Delta t_{\text{axle}} = \frac{4A\Omega}{c^2} \gamma^2 \quad \text{and} \quad \Delta\tau_{\text{det}} = \frac{4A\Omega}{c^2} \gamma$$

Here, $A = \pi R^2$, R , Ω , and $\beta = \Omega R/c$ are defined in the non-rotating inertial axle frame. The first expression is the coordinate return-time difference calculated in that frame. The second expression is the proper-time difference registered locally by the clock carried by the rotating detector.

In the intended low-velocity regime of the present derivation, $\beta \ll 1$ both expressions reduce at leading order to the familiar Sagnac result:

$$\Delta t_{\text{axle}} \approx \Delta\tau_{\text{det}} \approx \frac{4A\Omega}{c^2}$$

The distinction between coordinate return time and detector proper time becomes relevant only once relativistic higher-order factors are retained. At that point, the frame assignment can no longer remain implicit. The geometric correction series reconstructs the directed return-time asymmetry generated by the moving closure point in the axle frame. Special relativity then specifies how this already derived coordinate-time difference is read by the proper clock of the rotating detector. The proper-time conversion therefore preserves the geometric correction-series construction; it does not replace it and does not introduce a separate causal mechanism.

In the low-velocity limit $\gamma \rightarrow 1$, the axle-frame return-time result reduces to the familiar classical-kinematic Sagnac form. Retaining the distinction between axle-frame coordinate time and detector proper time yields the corresponding special-relativistic readout.

4. Phase as Readout, not Mechanism

In an interferometric implementation with continuous monochromatic light, the observed phase difference is the optical readout of the return-time mismatch derived above. The two propagation directions acquire different round-trip delays before their fields are recombined at the detector. When ω_{det} denotes the angular frequency referred to the detector's proper-time readout, the corresponding phase offset may be written as

$$\Delta\phi = \omega_{det}\Delta\tau_{det}.$$

For a pulsed implementation, the same underlying asymmetry would instead appear directly as two distinct return events on the detector's worldline. In either description, the observable is produced by the already established return-time mismatch. It should not be confused with the mechanism represented by the correction series: the series reconstructs the kinematic return-time asymmetry generated by the moving closure point, whereas the phase difference is its interferometric readout. In particular, the zig-zag convergence pattern of the series does not represent a physical oscillation, phase slippage, repeated frequency retuning, or any back-and-forth motion of the propagating light field.

5. Physical scope

The present derivation describes an idealised circular Sagnac configuration in flat spacetime. The optical path is assumed to form a prescribed closed loop of constant radius R , while the detector and the corresponding closure point rotate uniformly with angular velocity Ω . All quantities entering the geometric correction-series derivation, in particular R , $A = \pi R^2$, $L = 2\pi R$, $v = \Omega R$, and the coordinate return times t_+ and t_- are initially defined in the non-rotating inertial axle frame.

The geometric series converges whenever

$$|\beta| = \left| \frac{v}{c} \right| < 1$$

Within the idealised flat-spacetime model, the exact coordinate return-time difference is therefore

$$\Delta t_{axle} = \frac{4A\Omega}{c^2} \gamma^2$$

and the exact proper-time difference recorded by the co-rotating detector is

$$\Delta\tau_{det} = \frac{4A\Omega}{c^2} \gamma$$

These exact expressions do not require the approximation $v \ll c$. The low-velocity condition is required only when the familiar leading-order Sagnac form is used:

$$\Delta t_{axle} \approx \Delta\tau_{det} \approx \frac{4A\Omega}{c^2} \quad (\beta \ll 1)$$

The detector follows a circular worldline and is therefore accelerated even when v is small. This does not invalidate the proper-time conversion used above. In flat spacetime, for uniform circular motion at constant tangential speed v , the detector proper time remains related to the inertial axle-frame coordinate time by

$$d\tau_{\text{det}} = dt_{\text{axle}} \sqrt{1 - \beta^2}$$

What becomes physically problematic as v approaches c is not the mathematical return-time construction itself, but the realisation of a material rotating platform or optical apparatus with fixed radius and increasingly large tangential speed. Relativistic stresses, the nontrivial geometry of a rotating material ring, and the practical maintenance of a rigid or effectively fixed optical path would then have to be addressed separately. The present paper does not attempt to model such material or engineering constraints.

The scope of this derivation is also deliberately limited with respect to gravitational or background-rotational effects. The formulas derived here describe the Sagnac return-time asymmetry generated by a moving closure point on a rotating optical path in flat spacetime. They do not assume an aether, a refractive medium, a local variation of the speed of light, or a physically rotating spacetime as the cause of the ordinary Sagnac signal.

This limitation does not imply that additional rotational structures cannot be studied in other contexts. For example, a genuine background contribution such as frame dragging in a stationary axisymmetric spacetime would require an extended metric-based treatment containing the relevant off-diagonal rotational term. Such an extension would represent an additional physical contribution to the return-time asymmetry, not a reinterpretation of the elementary flat-spacetime Sagnac mechanism derived here.

Accordingly, the aim of the present derivation is not to replace standard engineering formulas or a full relativistic metric analysis. Its purpose is more specific: To show that, in the ordinary rotating-loop problem, the measured return-time difference can be reconstructed directly from the geometry of the moving closure condition; to distinguish this kinematic construction from the later interferometric phase readout and to keep axle-frame coordinate time separate from the proper-time reading of the rotating detector.